

EC-390: Digital Signal Processing

LECTURE NO 6

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Organization of Lecture

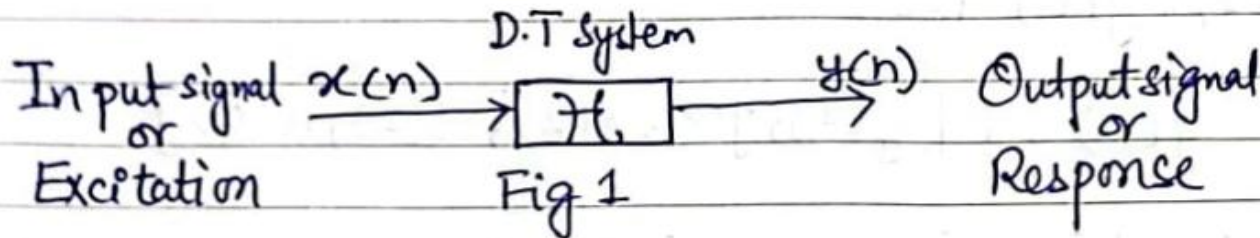
1. Discrete Time System.
 1. Mathematical Equation Governing Discrete Time System.
 2. Block Diagram Representation of Discrete Time System.
2. Response of LTI Discrete Time System in Time Domain.
 1. Zero-Input Response or Homogeneous Solution.
 2. Particular Solution.
 3. Zero-State Response.
 4. Total Response.

1. Discrete Time System

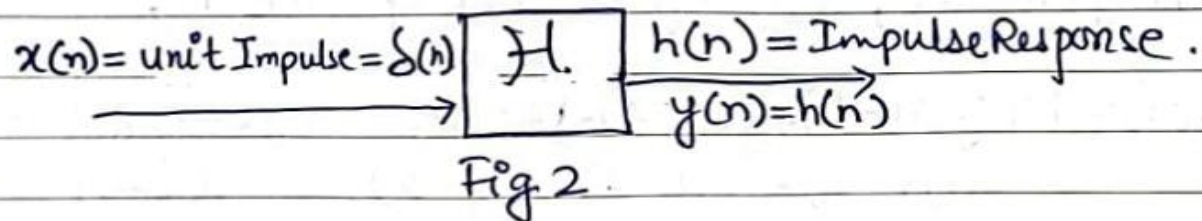
* A D.T system is a device or algorithm that operates on a D.T signal, called input or excitation, according to some well defined rule, to produce another D.T signal called the output or the response of the system.

$$\text{Response, } y(n) = H\{x(n)\}$$

where H denotes the transformation operator



* Impulse Response:



1. Discrete Time System

1. *Mathematical Equation Governing Discrete Time System.*

- * The response of a D.T system at any time instant depends on the present input, past inputs and past outputs.
- * Let us consider the response at $n=0$. let us assume a relaxed system and so at $n=0$, there is no past input or output. Therefore the response at $n=0$, is a function of present input alone i.e,

$$y(0) = F[x(0)]$$

1. Discrete Time System

1. *Mathematical Equation Governing Discrete Time System.*

* Let us consider the response at $n=1$. Now the present input is $x(1)$, the past input is $x(0)$ and past output $y(0)$. Therefore the response at $n=1$, is a function of $x(1)$, $x(0)$, $y(0)$ i.e.,

$$y(1) = F[y(0), x(1), x(0)]$$

* Let us consider the response at $n=2$. Now the present input is $x(2)$, the past inputs are $x(1)$ and $x(0)$, and past outputs are $y(1)$ and $y(0)$. Therefore the response at $n=2$ is a function of $x(2)$, $x(1)$, $x(0)$, $y(1)$, $y(0)$ i.e.,

$$y(2) = F[y(1), y(0), x(2), x(1), x(0)]$$

* Similarly at $n=3$, $y(3) = F(y(2), y(1), y(0), x(3), x(2), x(1), x(0))$, and so on.

1. Discrete Time System

1. *Mathematical Equation Governing Discrete Time System.*

* In general, at any instant of time n ,

$$y(n) = F[y(n-1), y(n-2), y(n-3), \dots, y(1), y(0), x(n), x(n-1), x(n-2), x(n-3), \dots, x(1), x(0)] \rightarrow \textcircled{1}$$

* When a D.T system satisfies the properties of linearity and time invariance then it is called an LTI (Linear Time Invariant) system. [We will discuss LTI in coming lectures]

* For an LTI system, the response $y(n]$ can be expressed as a weighted summation of dependent terms.

Equation $\textcircled{1}$ can be written as:

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) - a_3 y(n-3) - \dots + b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + b_3 x(n-3) + \dots \rightarrow \textcircled{2}$$

where $a_1, a_2, a_3, \dots$ and $b_1, b_2, b_3, \dots$ are constants

1. Discrete Time System

1. *Mathematical Equation Governing Discrete Time System.*

Note: Negative constants are inserted for output signals, because output signals are feedback from output to input. Positive constants are added/inserted for input signals, because input signals are feed forward from input to output

* Practically, the response $y(n)$ at any time instant n , may depend on 'N' number of past outputs, present input and 'M' number of past inputs where $M \leq N$.

Here equation (2) can be written as,

$$y(n) = -a_1 y(n-1) - a_2 y(n-2) - a_3 y(n-3) - \dots - a_N y(n-N) + b_0 x(n) + b_1 x(n-1) + b_2 x(n-2) + b_3 x(n-3) + \dots + b_M x(n-M)$$

$$y(n) = -\sum_{m=1}^N a_m y(n-m) + \sum_{m=0}^M b_m x(n-m) \rightarrow (3)$$

1. Discrete Time System

1. *Mathematical Equation Governing Discrete Time System.*

* The equation (3) is a constant coefficient difference equation, governing the input-output relation of an LTI discrete time system.

if $N=1$, the D.T system is called 1st order system.

if $N=2$, " " " " " 2nd " " .

if $N=3$, " " " " " 3rd " " .

* The general difference equation governing 1st order D.T LTI system is:

$$y(n) = -a_1 y(n-1) + b_0 x(n) + b_1 x(n-1)$$

* The general difference equation governing 2nd order D.T LTI system is:

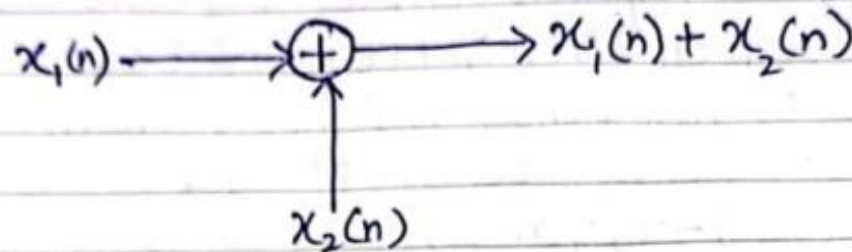
$$y(n) = -a_2 y(n-2) - a_1 y(n-1) + b_0 x(n) + b_1 x(n-1) + b_2 x(n-2)$$

1. Discrete Time System

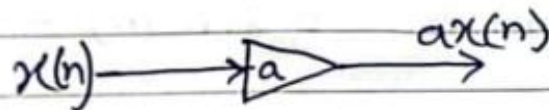
2. Block Diagram Representation of Discrete Time System.

* The D.T system can be represented diagrammatically by block diagram. These diagrammatic representations are useful for physical implementation of D.T system in hardware or software.

i) Adder



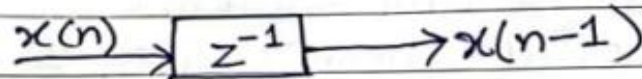
ii) Constant Multiplier



1. Discrete Time System

2. Block Diagram Representation of Discrete Time System.

iii) Unit delay element



iv) Unit advance element

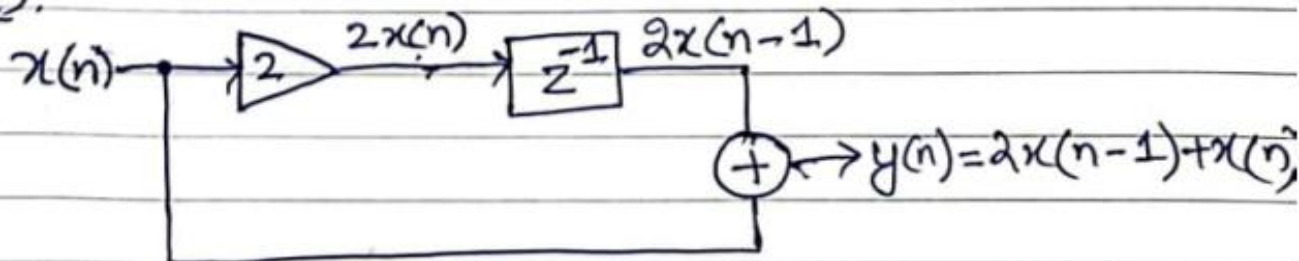


Example 1: Construct block diagram of the D.T systems whose input-output relations are described by the following difference equations.

a) $y(n] = 2x(n-1) + x(n)$

b) $y(n] = 0.4y(n-1) + x(n) - 3x(n-2)$

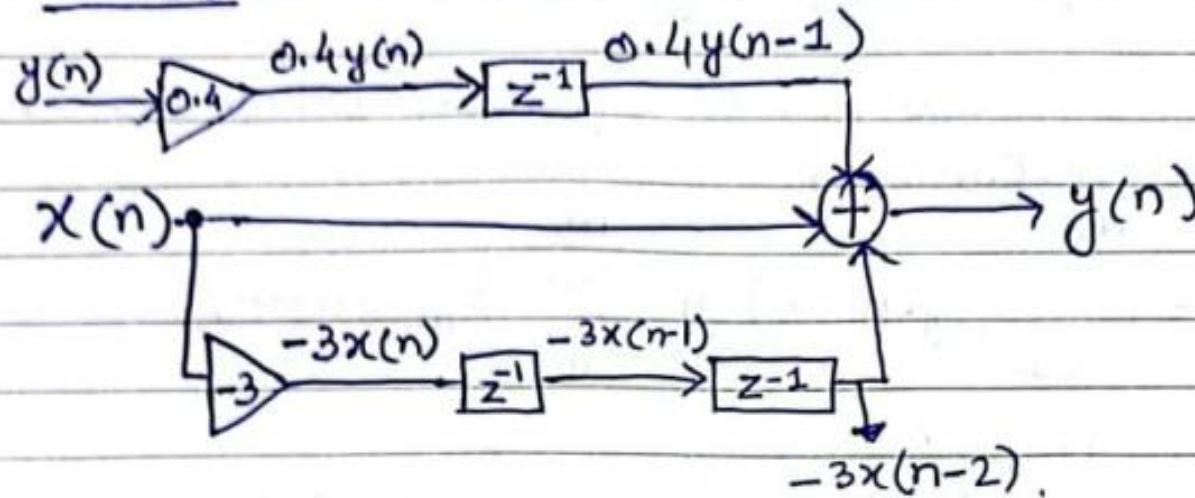
Solⁿ(a):



1. Discrete Time System

2. Block Diagram Representation of Discrete Time System.

Solⁿ(b):



Example 2: (H.W.)

Construct block diagrams for the following difference equations.

a) $y(n] = 0.5x(n] + 0.5x(n-1]$

b) $y(n] = 0.25y(n-1] + x(n] - 2x(n-2]$

2. Response of LTI Discrete Time System in Time Domain

* Equation (3) is the general equation governing an LTI D.T system i.e.,

$$y(n) = - \sum_{m=1}^N a_m y(n-m) + \sum_{m=0}^M b_m x(n-m)$$

$$\underbrace{y(n) + \sum_{m=1}^N a_m y(n-m)}_{\text{L.H.S}} = \underbrace{\sum_{m=0}^M b_m x(n-m)}_{\text{R.H.S}} \rightarrow (4)$$

2. Response of LTI Discrete Time System in Time Domain

$$\text{L.H.S} = \sum_{m=0}^N a_m y(n-m)$$

$$y(n) + \sum_{m=1}^N a_m y(n-m) = \sum_{m=0}^N a_m y(n-m)$$

$$y(n) + a_1 y(n-1) + a_2 y(n-2) + \dots + a_N y(n-N) =$$

$$a_0 y(n) + a_1 y(n-1) + a_2 y(n-2) + \dots + a_N y(n-N)$$

* If we substitute $a_0 = 1$ in above equation L.H.S = R.H.S

* Therefore eq(4) can be written as:

$$\sum_{m=0}^N \overbrace{a_m y(n-m)}^{\text{Output}} = \sum_{m=0}^M \overbrace{b_m x(n-m)}^{\text{Input}} \text{ with } a_0 = 1 \rightarrow \textcircled{5}$$

2. Response of LTI Discrete Time System in Time Domain

* Equation (5) is the response $y(n)$ of LTI system, which consists of two parts. In mathematics, the two parts of the solution or response $y(n)$ are homogeneous solution $y_h(n)$ and particular solution $y_p(n)$.

$$\therefore \text{Response, } y(n) = y_h(n) + y_p(n) \rightarrow (6)$$

* The homogeneous solution is the response of the system when there is no input. The particular solution $y_p(n)$ is the solution of difference equation for specific input signal $x(n)$ for $n \geq 0$.

2. Response of LTI Discrete Time System in Time Domain

* In signals and systems, two parts of the solution $y(n)$ are called zero-input response $y_{zi}(n)$ and zero-state response $y_{zs}(n)$.

$$\therefore \text{Response, } y(n) = y_{zi}(n) + y_{zs}(n) \rightarrow (7)$$

* The zero-input response is mainly due to initial conditions (or initial stored energy) in the system. Hence zero-input response is also called free response or natural response. The zero-input response is given by homogeneous solution with constants evaluated using initial conditions.

* The zero-state response is the response of the system due to input signal and with zero initial condition. Hence the zero-state response is called forced response. The zero-state response or forced response is given by the sum of homogeneous solution and particular solution with zero initial conditions.

2. Response of LTI Discrete Time System in Time Domain

1. Zero-Input Response or Homogeneous Solution.

* The zero-input response is obtained from homogeneous solution $y_h(n)$ with constants evaluated using initial condition.

$\therefore$ Zero-input response, $y_{zi}(n) = y_h(n)$ with constants evaluated using initial conditions.

* The homogeneous solution is obtained when $x(n) = 0$.

Eq (5)

$$\sum_{m=0}^N a_m y(n-m) = 0 \rightarrow (8)$$

* Let us assume that the solution of eq (8) is in the form of an exponential i.e.,

$$y(n) = \lambda^n$$

also, $y(n-m) = \lambda^{n-m}$

Substitute $y(n-m) = \lambda^{n-m}$ in eq (8) we get

2. Response of LTI Discrete Time System in Time Domain

1. Zero-Input Response or Homogeneous Solution.

$$\sum_{m=0}^N a_m \lambda^{n-m} = 0$$

* On expanding the above equation (by taking $a_0=1$), we get:

$$a_0 \lambda^n + a_1 \lambda^{n-1} + a_2 \lambda^{n-2} + \dots + a_{N-1} \lambda^{n-(N-1)} + a_N \lambda^{n-N} = 0$$
$$\lambda^{n-N} (\lambda^N + a_1 \lambda^{N-1} + a_2 \lambda^{N-2} + \dots + a_{N-1} \lambda + a_N) = 0$$

* Now the characteristic polynomial of the system is given by,

$$\lambda^N + a_1 \lambda^{N-1} + a_2 \lambda^{N-2} + \dots + a_{N-1} \lambda + a_N = 0$$

The characteristic polynomial has N roots, which are denoted by/as: $\lambda_1, \lambda_2, \dots, \lambda_N$.

* The roots of the characteristic polynomial may be distinct real roots, repeated real roots or complex roots. The assumed solutions for various types of roots are given below:

2. Response of LTI Discrete Time System in Time Domain

1. Zero-Input Response or Homogeneous Solution.

Distinct Real Roots:

* Let the roots $\lambda_1, \lambda_2, \lambda_3, \dots, \lambda_N$ be distinct real roots. Now the homogeneous solution will be in the form,

$$y_h(n) = C_1 \lambda_1^n + C_2 \lambda_2^n + C_3 \lambda_3^n + \dots + C_N \lambda_N^n$$

where $C_1, C_2, C_3, \dots, C_N$ are constants that can be evaluated using initial conditions.

Repeated Real Roots:

* Let one of the real roots λ_1 repeats 'p' times and the remaining $(N-p)$ roots are distinct real roots. Now, the homogeneous solution is in the form,

$$y_h(n) = (C_1 + C_2 n + C_3 n^2 + \dots + C_p n^{p-1}) \lambda_1^n + C_{p+1} \lambda_{p+1}^n + \dots + C_N \lambda_N^n$$

where $C_1, C_2, C_3, \dots, C_N$ are constants that can be evaluated using initial conditions.

2. Response of LTI Discrete Time System in Time Domain

1. Zero-Input Response or Homogeneous Solution.

Complex Roots:

* Let the characteristic polynomial has a pair of complex roots λ and λ^* and the remaining $(N-2)$ roots be distinct real roots. Now the homogeneous solution will be in the form,

$$y(n) = r^n [C_1 \cosh n\theta + C_2 \sinh n\theta] + C_3 \lambda_3^n + C_4 \lambda_4^n + \dots + C_N \lambda_N^n$$

where, $\lambda = a + jb$, $\lambda^* = a - jb$, $r = \sqrt{a^2 + b^2}$, $\theta = \tan^{-1}\left(\frac{b}{a}\right)$

$C_1, C_2, C_3, \dots, C_N$ are constants that can be evaluated using initial conditions.

2. Response of LTI Discrete Time System in Time Domain

2. Particular Solution.

* The particular solution, $y_p(n)$ is the solution of the difference equation for specific input signal $x(n)$ for $n \geq 0$. Since the input signal may have different form, the particular solution depends on the form or type of the input signal $x(n)$.

- If $x(n)$ is constant, then $y_p(n)$ is also a constant.

For example:

Let, $x(n) = u(n)$; now, $y_p(n) = K u(n)$

- If $x(n)$ is exponential, then $y_p(n)$ is also an exponential.

For example:

Let, $x(n) = a^n u(n)$; now $y_p(n) = K a^n u(n)$

- If $x(n)$ is sinusoid, then $y_p(n)$ is also a sinusoid.

For example:

Let, $x(n) = A \cos \omega_0 n$; now $y_p(n) = K_1 \cos \omega_0 n + K_2 \sin \omega_0 n$

2. Response of LTI Discrete Time System in Time Domain

2. Particular Solution.

Particular Solution:

The general form of particular solution for various types of inputs are listed in table below.

Input signal, $x(n)$	Particular Solution, $y_p(n)$
A	$\rightarrow K$
$A B^n$	$\rightarrow K B^n$
$A n^B$	$\rightarrow K_0 n^B + K_1 n^{(B-1)} + \dots + K_B$
$A n^B B^n$	$\rightarrow A n^B (K_0 n^B + K_1 n^{(B-1)} + \dots + K_B)$
$\left. \begin{matrix} A \cos \omega_0 n \\ A \sin \omega_0 n \end{matrix} \right\}$	$\rightarrow K_1 \cos \omega_0 n + K_2 \sin \omega_0 n$

2. Response of LTI Discrete Time System in Time Domain

3. Zero-State Response.

* The Zero-state response or forced response is obtained from the sum of homogeneous solution and particular solution and evaluating the constants with zero initial conditions.

$$\therefore y_{zs}(n) = y_h(n) + y_p(n)$$

with constants $C_1, C_2, \dots, C_N$ evaluated with zero initial conditions.

2. Response of LTI Discrete Time System in Time Domain

4. Total Response.

* The total response is given by sum of homogeneous solution and particular solution.

$$\therefore y(n) = y_h(n) + y_p(n)$$

2. Response of LTI Discrete Time System in Time Domain

4. Total Response.

Procedure to Determine Total Response:

1. Determine the homogeneous solution $y_h(n)$ with constants $C_1, C_2, \dots, C_N$.
2. Determine the particular solution $y_p(n)$ and evaluate the constants K for any value of $n \geq 1$ so that no term vanishes.
3. Now the total response is given by the sum of $y_h(n)$ and $y_p(n)$ i.e.,
$$y(n) = y_h(n) + y_p(n)$$
4. The total response will have N number of constants $C_1, C_2, C_3, \dots, C_N$. Evaluate the given equation and the total response for $n = 0, 1, 2, \dots, N-1$ and form two sets of N number of equations and solve the constants $C_1, C_2, \dots, C_N$.

2. Response of LTI Discrete Time System in Time Domain

Example 3: Determine the response of first order D.T system governed by the difference equation.

$$y(n) = -0.5y(n-1) + x(n)$$

when the input is unit step and with initial conditions:

a) $y(-1) = 0$

b) $y(-1) = 1/3$.

2. Response of LTI Discrete Time System in Time Domain

Solution:

Given that, $y(n) = -0.5y(n-1) + x(n)$

$$\therefore y(n) + 0.5y(n-1) = x(n) \longrightarrow \textcircled{1}$$

Step 1: Homogeneous solution:

put $x(n) = 0$ in eq (1)

$$\therefore y(n) + 0.5y(n-1) = 0 \longrightarrow \textcircled{2}$$

put $y(n) = \lambda^n$ in eq (2)

$$\lambda^n + 0.5\lambda^{(n-1)} = 0$$

$$\lambda^{n-1}(\lambda + 0.5) = 0$$

$$\lambda = -0.5$$

2. Response of LTI Discrete Time System in Time Domain

The homogeneous solution $y_h(n)$ is given by:

$$y_h(n) = C\lambda^n$$

$$y_h(n) = C(-0.5)^n; \text{ for } n \geq 0 \rightarrow \textcircled{3}$$

Step 2: Particular Solution

Given that the input is unit step and so the particular solution will be in the form,

$$y(n) = Ku(n) \rightarrow \textcircled{4}$$

on substituting for $y(n)$ from equation (4) in equation (1) we get

$$y(n) + 0.5y(n-1) = x(n)$$

$$Ku(n) + 0.5Ku(n-1) = u(n) \rightarrow \textcircled{5}$$

2. Response of LTI Discrete Time System in Time Domain

In order to determine the value of 'K' in eq(5).
let us evaluate it for $n=1$ ($\because$ we have to evaluate eq(5) for any $n \geq 1$ such that none of the term vanishes).

eq(5):

$$K u(1) + 0.5 K u(1-1) = u(1)$$

$$K u(1) + 0.5 K u(0) = u(1)$$

$$K \cdot 1 + 0.5 K \cdot 1 = 1$$

$$K + 0.5 K = 1$$

$$1.5 K = 1$$

$$K = \frac{1}{1.5} = \frac{2}{3}$$

The particular solution $y_p(n)$ is given by:

$$y_p(n) = K u(n)$$

$$= \frac{2}{3} u(n) ; \text{ for all } n$$

$$y_p(n) = \frac{2}{3} ; \text{ for } n \geq 0$$

2. Response of LTI Discrete Time System in Time Domain

Step 3 and Step 4: The total response is given by:

$$y(n) = y_h(n) + y_p(n)$$

$$y(n) = C(-0.5)^n + \frac{2}{3} ; \text{ for } n \geq 0 \rightarrow \textcircled{6}$$

2. Response of LTI Discrete Time System in Time Domain

At $n=0$ in given equation (1) we get

$$y(n) + 0.5y(n-1) = x(n)$$

$$y(n) = x(n) - 0.5y(n-1)$$

$$y(0) = x(0) - 0.5y(-1)$$

$$y(0) = 1 - 0.5y(-1) \rightarrow \textcircled{7}$$

At $n=0$ in equation (6) (homogeneous equation) we get

$$y(n) = C(-0.5)^n + \frac{2}{3}$$

$$y(0) = C(-0.5)^0 + \frac{2}{3}$$

$$y(0) = C + \frac{2}{3} \rightarrow \textcircled{8}$$

Comparing (7) and (8) we get

$$C + \frac{2}{3} = 1 - 0.5y(-1)$$

$$C = 1 - 0.5y(-1) - \frac{2}{3}$$

$$= 1 - \frac{2}{3} - 0.5y(-1)$$

$$C = \frac{1}{3} - 0.5y(-1) \rightarrow \textcircled{9}$$

2. Response of LTI Discrete Time System in Time Domain

put value of 'C' in eq (6) we get .

$$y(n) = C(-0.5)^n + \frac{2}{3}$$

$$y(n) = \left[\frac{1}{3} - 0.5y(-1) \right] (-0.5)^n + \frac{2}{3} \rightarrow (10)$$

a) when $y(-1) = 0$; eq (10) will be

$$y(n) = \frac{1}{3}(-0.5)^n + \frac{2}{3} \text{ for } n \geq 0$$

b) when $y(-1) = 1/3$; eq (10) will be

$$y(n) = \left[\frac{1}{3} - 0.5\left(\frac{1}{3}\right) \right] (-0.5)^n + \frac{2}{3}$$

$$y(n) = \frac{1}{6}(-0.5)^n + \frac{2}{3} ; \text{ for } n \geq 0$$

2. Response of LTI Discrete Time System in Time Domain

Example 4 (H.W)

Determine the response $y(n]$, $n \geq 0$ of the system described by the second order difference equation.

$$y(n) - 2y(n-1) - 3y(n-2) = x(n) + 4x(n-1)$$

when the input signal is, $x(n) = 2^n u(n)$ and with initial conditions:

a) $y(-2) = 0$

b) $y(-1) = 5$

References: Textbooks

1. *“Discrete-Time Signal Processing”, International Edition, by Oppenheim, Alan V; Schafer, Ronald W, (2013), Pearson.*
2. *“Digital Signal Processing Using Matlab”, 4th Edition by Vinay K. Ingle, John G. Proakis, (2016), Cengage Learning.*
3. *Related material from open source, mainly internet.*